Deterministic Modeling - ODE Exercises

Matrix Euler Scheme & Stability Analysis

October 2026

Convergence Condition for Fixed-Point Iteration

When solving non-linear initial value problems $y' = f(y)$ using the implicit matrix Euler scheme:

$$D\mathbf{y}^{(k+1)} = \mathbf{b}(\mathbf{y}^{(k)})$$

we solve the system iteratively using a simple fixed-point method. For each node i, the iterative step can be written as:

$$y_i^{(k+1)} = y_{i-1}^{(k+1)} + h \cdot f\left(y_i^{(k)}\right)$$

For this simple iteration to converge at a given node, the magnitude of the local change must not grow between iterations. This leads to a **necessary condition** for the convergence of fixed-point iteration:

$$|h \cdot f'(y)| < 1 \quad \Longrightarrow \quad h < \frac{1}{|f'(y)|}$$

Key Takeaways:

1. **Necessary Condition:** The inequality $h < 1/|f'(y)|$ is a **necessary condition** for the simple fixed-point iteration to converge. If this condition is violated ($h \geq 1/|f'(y)|$), the iterative process will diverge and shoot off to $\pm\infty$.
2. **Dependence on State y:** For non-linear differential equations, $f'(y)$ depends on the current value of y. If an initial condition y_0 leads the solution into a region where $|f'(y)|$ is large, you must decrease the step size h (increase N) to keep $h \cdot |f'(y)| < 1$.
3. **Distinction:** Remember that this limitation comes from using simple fixed-point iteration to solve the non-linear matrix equation, not from the implicit Euler scheme itself.

Important Implementation Note

Crucial Rule for the Iterative Solver: When using the matrix differentiation operator D of dimension $(N+1) \times (N+1)$ (where N is the number of grid intervals), you **must** manually ensure that the first element of the right-hand side vector $\mathbf{b}$ is explicitly set and preserved as the initial condition ($\mathbf{b}[0] = y_0$) inside every iteration loop.

Because the first row of matrix D encodes the boundary/initial condition ($1 \cdot y_0 = y_0$), overwriting $\mathbf{b}[0]$ with $f(y_0)$ will invalidate the initial condition, leading to incorrect system boundaries and causing the numerical solver to fail or crash.

Exercise List

1. **Task 1: Malthus Growth Model (Exponential Growth)**

   - **Equation:**
   $$\frac{dy}{dt} = ry$$
   - **Initial Condition:** $y(0) = 1$ (or $y_0 = 1$)
   - **Parameters:** $r = 1.0$, time interval $t \in [0, 3]$, step count $N = 30$ (step size $h = 3/N = 0.1$).
   - **Analytical Solution:** $y(t) = e^t$
   - **Goal:** Implement the matrix Euler scheme, plot the numerical solution against the exact solution, and compute the error metrics e_1, e_2, e_3, and RMSE (see Error Definitions below).

2. **Task 2: Verhulst Model (Logistic Growth)**

   - **Equation:**
   $$\frac{dy}{dt} = ry\left(1 - \frac{y}{K}\right)$$
   - **Part A (Steady States):** Set $r = 0.1$, $K = 1$. Simulate the system for initial values:
   $$y_0 \in \{0, 0.25, 0.5, 0.75, 1, 1.5\}$$
   Observe the stability of stationary states $y^* = 0$ (unstable) and $y^* = K$ (stable).
   - **Part B (Error Analysis):** For $y_0 = 0.5$, compare the numerical solution against the analytical solution $y(t) = \frac{K}{1+(K/y_0-1)e^{-rt}}$ by calculating the error metrics e_1, e_2, and RMSE across varying step sizes $h \in \{0.1, 0.01, 0.001\}$.
   - *Note:* For $y_0 > 1$, a smaller step size h (larger N) is required to satisfy the contraction mapping condition $h < 1/|f'(y)|$ and prevent divergence.

3. **Task 3: Gompertz Growth Model**

   - **Equation:**
   $$\frac{dy}{dt} = -ry\ln\left(\frac{y}{K}\right)$$
   - **Parameters:** $K = 1.0$, $r = 0.5$, $y_0 = 0.1$, time interval $t \in [0, 10]$, step count $N = 100$ ($h = 0.1$).
   - **Analytical Solution:** $y(t) = K \cdot \exp\left(\ln\left(\frac{y_0}{K}\right)e^{-rt}\right)$
   - **Goal:** Evaluate the performance of the fixed-point matrix iteration under logarithmic nonlinearity and compute all error metrics (e_1, e_2, e_3, RMSE) relative to the analytical solution.

4. **Task 4: Budworm Population Dynamics (Hysteresis Model)**

   - **Equation:**
   $$\frac{dy}{dt} = ry\left(1 - \frac{y}{8}\right) - \frac{10y^2}{1+y^2}$$
   - **Part A (Bistability Region):** Propose parameters $r_1, r_2 > 0$ ($r_2 > r_1$) such that $f(y) = 0$ has 4 roots for r_1 and 2 roots for r_2. First, verify the stability of stationary states for r_1 on a short time domain up to $T = 2$ using a sufficiently small step size (e.g., $h = 0.01$) across initial values $y_0 \in [0, 6]$.

- **Part B (Dynamic Hysteresis):** For $y_0 = 0.25$ and $h = 0.01$, simulate the system with a time-varying growth rate $r(t)$:

$$r(t) = \begin{cases} r_1, & 0 \le t \le 3 \\ r_2, & 3 < t \le 6 \\ r_1, & 6 < t \le 9 \end{cases}$$

  Demonstrate that the solution path $y(t)$ exhibits hysteresis (memory effect).

- **Part C (Double-Step Error Estimation):** Since no simple analytical solution exists, estimate the global error at terminal time $T = 5$ for $r = 2, y_0 = 0.25$ using the double-step method by comparing steps $h = 0.01$ and $2h = 0.02$:

$$e_h \approx |\hat{y}_h(T) - \hat{y}_{2h}(T)|$$

Error Definitions

Let $y(t_i)$ denote the exact analytical solution at node t_i, and $\hat{y}_i$ be the numerical approximation calculated by the scheme for $i = 0, 1, \ldots, N$. The global numerical error metrics are defined as follows:

- **Final Point Error (e_1):** Measures the absolute deviation solely at the terminal time $T = t_N$:

$$e_1 = |y(T) - \hat{y}_N|$$

- **Maximum Absolute Error / ℓ^∞-norm (e_2):** Measures the worst-case pointwise deviation across the entire time grid:

$$e_2 = \|\mathbf{y} - \hat{\mathbf{y}}\|_\infty = \max_{0 \le i \le N} |y(t_i) - \hat{y}_i|$$

- **Euclidean Error Norm / ℓ^2-norm (e_3):** Quantifies the cumulative magnitude of pointwise differences across all nodes:

$$e_3 = \|\mathbf{y} - \hat{\mathbf{y}}\|_2 = \left(\sum_{i=0}^{N} (y(t_i) - \hat{y}_i)^2 \right)^{1/2}$$

- **Root Mean Square Error (RMSE):** Represents the average deviation per grid point, normalizing e_3 by the number of steps:

$$\text{RMSE} = \sqrt{\frac{1}{N+1} \sum_{i=0}^{N} (y(t_i) - \hat{y}_i)^2} = \frac{e_3}{\sqrt{N+1}}$$